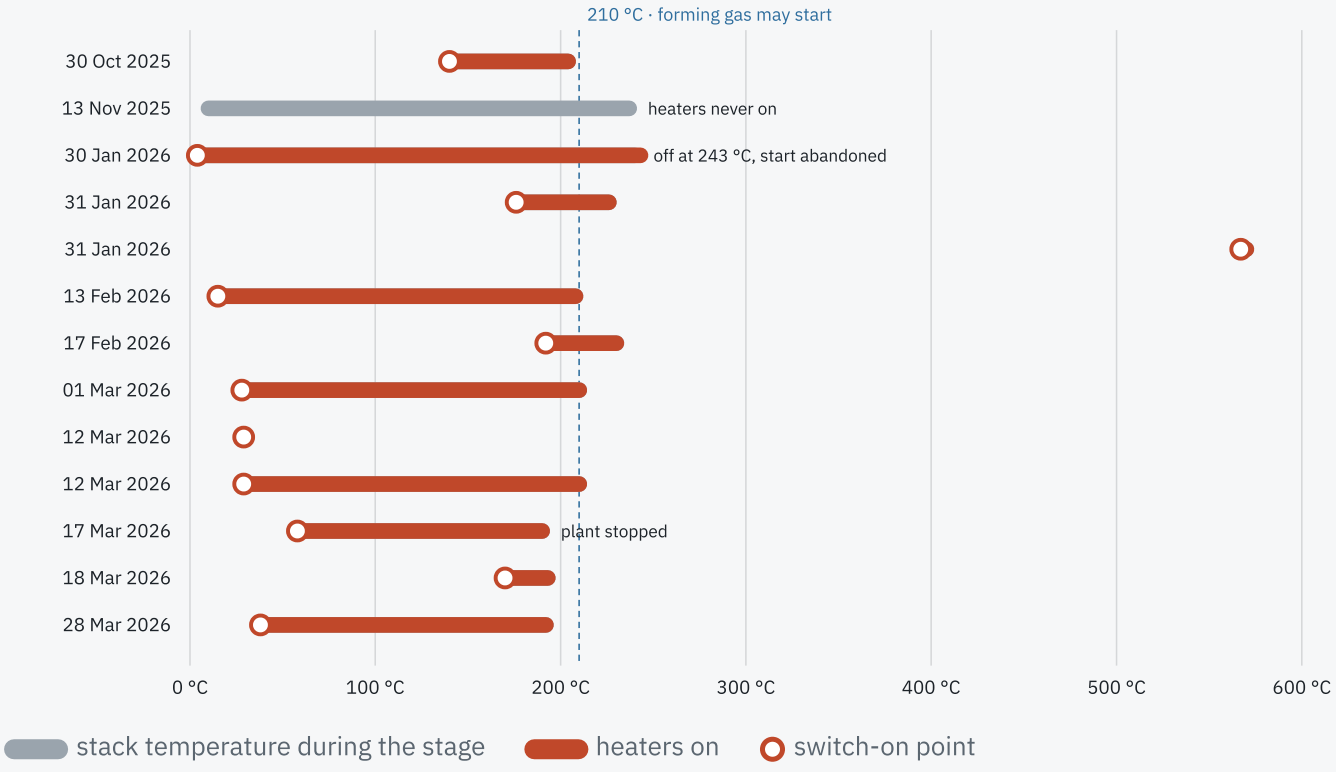
HEAT-UP AIR STAGE · 14 RUNS · OCT 2025 – MAR 2026

The heaters follow the air flow, not the stack temperature

In heat-up air, five heaters switch on within a few minutes of the air starting, at whatever temperature the stack happens to be. They are the reformer heater (ERH), the gas heater (EAGH), the off-gas heaters (EAOH 1 and 2) and the recycle heater (EAOH 3). Once on, they stay on until the stage ends. We found no stack temperature at which they switch on or off.

Why it matters: a stage rule or a data generator should switch these heaters on with the air, not at a stack temperature.

What each run looked like



Each row is one heat-up-air run. The heaters came on at stack temperatures from 4 °C to 567 °C and stayed on to the end of the stage, past the 210 °C line where forming gas can start. The 22 Mar 2026 run is left out because its stack temperature reading was frozen. Source: 39-day plant log, 1-minute averages. A heater counts as “on” when its current is above 0.5 A.

Three things the data shows

On: about 2 minutes after the air starts

The median wait is 2–3 minutes after the air starts, and the range is 0–9 minutes. The stack temperature at switch-on ranged from 4 °C to 567 °C. All five heaters come on together, within a minute or two of each other.

Off: never set by temperature

The heaters came on in 12 of the 13 usable runs. In 10 of those they ran to the end of the stage, usually carrying on into heat-up N_2+H_2. They switched off early only twice, both times because the start was being stopped (at 243 °C and 210 °C).

Power: steady, not adjusted to temperature

The current stays flat as the stack warms (correlation with stack temperature about 0). ERH, EAGH and EAOH 1·2 draw about 3.3 A and EAOH 3 about 8.4 A. In mid-March they drew more, about 5 A and 13 A.

> **If you remember one thing:** in heat-up air, the five heaters switch on with the air and stay on. Stack temperature does not switch them. See every run →

RUN BY RUN

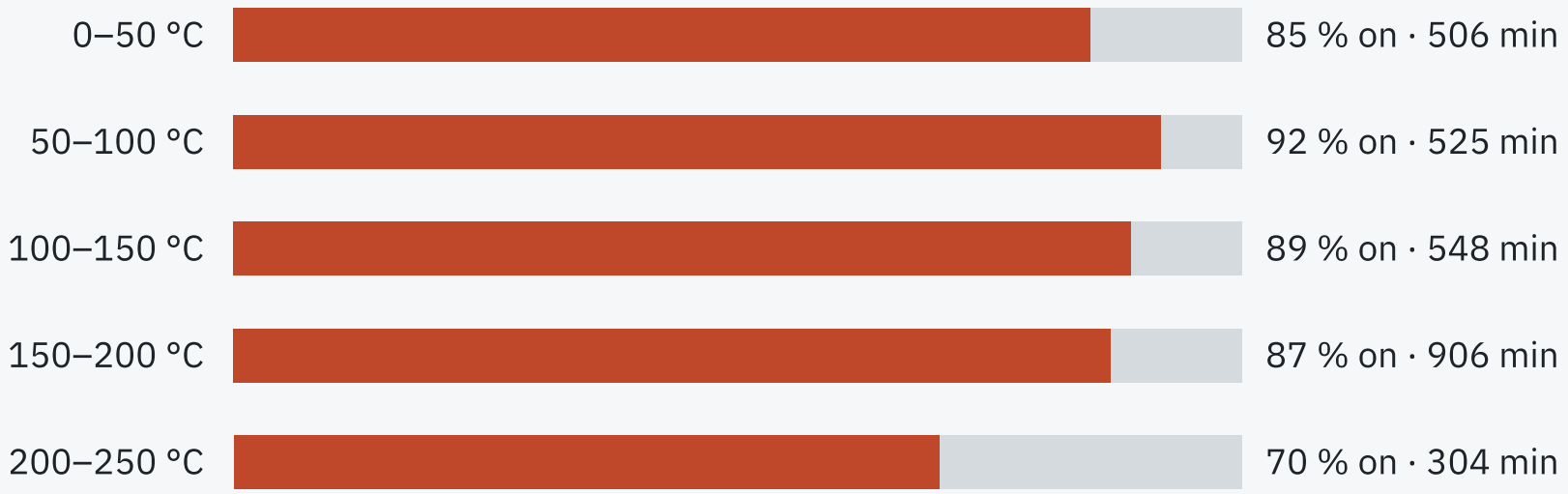
Every heat-up-air run, with stack temperature

“Switch-on” is the number of minutes after heat-up air began until ERH, EAGH and EAOH 1·2 were all on, with the stack temperature at that moment. EAOH 3 has its own column because it behaved differently once. “On share” is the part of the stage the heaters were on. Outside the runs with a note, the only off time is the few minutes before switch-on. The summary chart shows the same runs.

START	MIN	STACK °C START → END	ERH · EAGH · EAOH 1·2 ON AFTER	EAOH 3 ON AFTER	ON SHARE	NEXT STAGE	NOTE
30 Oct 2025 11:29	169	139 → 204	2 min · 140 °C	2 min	98 %	Heat-up N_2+H_2	
13 Nov 2025 10:47	330	10 → 237	never	never	0 %	Heat-up N_2+H_2	Heaters never came on; the air heater alone warmed the stack
30 Jan 2026 10:10	540	4 → 240	0 min · 4 °C	0 min	94 %	Air-flow cooling	All heaters switched off at 243 °C; start abandoned, went to air-flow cooling
31 Jan 2026 08:09	120	176 → 226	3 min · 176 °C	1 min	98 %	Heat-up N_2+H_2	
31 Jan 2026 17:24	14	567 → 570	2 min · 567 °C	0 min	92 %	Stopped	Short 14-min dip back to heat-up air while already hot

START	MIN	STACK °C START → END	ERH · EAGH · EAOH 1·2 ON AFTER	EAOH 3 ON AFTER	ON SHARE	NEXT STAGE	NOTE
13 Feb 2026 14:32	278	15 → 208	1 min · 15 °C	0 min	99 %	Heat-up N_2+H_2	
17 Feb 2026 16:56	107	192 → 230	3 min · 192 °C	1 min	98 %	Heat-up N_2+H_2	
01 Mar 2026 09:05	297	28 → 210	3 min · 28 °C	3 min	98 %	Heat-up N_2+H_2	
12 Mar 2026 10:29	64	28 → 29	4 min · 29 °C	3 min	93 %	Stopped	Short 64-min try; plant stopped, restarted at 11:34
12 Mar 2026 11:34	232	29 → 210	0 min · 29 °C	0 min	100 %	Heat-up N_2+H_2	
17 Mar 2026 14:56	312	58 → 190	3 min · 58 °C	2 min	98 %	Stopped	Heaters, air and logging all stopped together at 210 °C (plant stop)
18 Mar 2026 07:06	99	169 → 193	9 min · 170 °C	8 min	90 %	Heat-up N_2+H_2	EAOH 3 came on, then went off after 13 min at 170 °C; the others stayed on
22 Mar 2026 00:00	573	274 → 274	never	never	0 %	Heat-up N_2+H_2	Stack temperature frozen at 274 °C for 9.5 h (sensor or log fault); heaters off; left out
28 Mar 2026 11:45	241	37 → 192	6 min · 38 °C	5 min	97 %	Heat-up N_2+H_2	

Share of minutes with heaters on, by stack temperature



Heat-up-air minutes grouped by stack temperature. There is no step at any temperature. Most of the off time comes from the 13 Nov 2025 run, when the heaters never came on. The 200–250 °C band is lower because it also holds the minutes after the 30 Jan heaters were switched off. The 250–300 °C band is left out because it is almost all the frozen-sensor run of 22 Mar.

> **If you remember one thing:** both switch-offs happened because the start was being stopped. Neither was set by temperature.